

behaviors like multiple choice reasoning (Lieberum et al., 2023), indirect object identification (Wang et al., 2023), or computing operations (Hanna et al., 2023). Other examples include analyzing circuits for Python docstrings (Heimersheim & Jett, 2023), "an" vs "a" usage (Miller & Neo, 2023), and price tagging (Wu et al., 2023b). Case studies often construct datasets using templates filled by placeholder values to enable precise control for causal interventions (Wang et al., 2023; Hanna et al., 2023; Wu et al., 2023b).

Toy Models – Comprehensive Analysis. Small models trained on specialized mathematical or algorithmic tasks enable more comprehensive reverse engineering of learned algorithms (Nanda et al., 2023a; Zhong et al., 2023; Chughtai et al., 2023). Even simple arithmetic operations can involve complex strategies and multiple algorithmic solutions (Nanda et al., 2023a; Zhong et al., 2023). Characterizing these algorithms helps test hypotheses around generalizable mechanisms like variable binding (Feng & Steinhardt, 2023; Davies et al., 2023) and arithmetic reasoning (Stolfo et al., 2023). The work by Varma et al. (2023) builds on the work that analyzes transformers trained on modular addition (Nanda et al., 2023a) and explains *grokking* in terms of circuit efficiency, illustrating how a comprehensive understanding of a toy model can enable interesting analyses on top of that understanding.

Towards Universality. The ultimate goal is to uncover general principles that transfer across models and tasks, such as induction heads for in-context learning (Olsson et al., 2022), variable binding mechanisms (Feng & Steinhardt, 2023; Davies et al., 2023), arithmetic reasoning (Stolfo et al., 2023; Brinkmann et al., 2024), or retrieval tasks (Variengien & Winsor, 2023). Despite promising results, debates surround the *universality* hypothesis – the idea that different models learn similar features and circuits when trained on similar tasks. (Chughtai et al., 2023) finds mixed evidence for universality in group composition, suggesting that while families of circuits and features can be characterized, precise circuits and development order may be arbitrary.

Towards High-level Mechanisms. Causal interventions can extract a high-level understanding of computations and representations learned by large language models (Variengien & Winsor, 2023; Hendel et al., 2023; Feng & Steinhardt, 2023; Zou et al., 2023). Recent work focuses on intervening in internal representations to study high-level concepts and computations encoded. For example, Hendel et al. (2023) patched residual stream vectors to transfer task representations, while Feng & Steinhardt (2023) intervened on residual streams to argue that models generate IDs to bind entities to attributes. Techniques for *representation engineering* (Zou et al., 2023) extract reading vectors from model activations to stimulate or inhibit specific concepts. Although these interventions don’t operate via specific mechanisms, they offer a promising approach for extracting high-level causal understanding and bridging bottom-up and top-down interpretability approaches.

5.4 Automation: Scaling Post-Hoc Interpretability

As models become more complex, automating key aspects of the interpretability workflow becomes increasingly crucial. Tracing a model’s computational pathways is highly labor-intensive, quickly becoming infeasible as the model size increases. Automating the discovery of relevant circuits and their functional interpretation represents a pivotal step towards scalable and comprehensive model understanding (Nainani, 2024).

Dissecting Models into Interpretable Circuits. The first major automation challenge is identifying the critical computational sub-circuits or components underpinning a model’s behavior for a given task. A pioneering line of work aims to achieve this via efficient **masking** or **patching** procedures. Methods like *automated circuit discovery* (Conmy et al., 2023) and *attribution patching* (Syed et al., 2023; Kramár et al., 2024) iteratively knock out model activations, pinpointing components whose removal has the most significant impact on performance. This masking approach has proven scalable even to large models (Lieberum et al., 2023).

Other techniques take a more top-down approach. Davies et al. (2023) specify high-level causal properties (desiderata) that components solving a target subtask should satisfy and then learn binary masks to expose those component subsets. Ferrando & Voita (2024) construct *information flow graphs* highlighting key nodes

and operations by tracing attribution flows, enabling extraction of general information routing patterns across prediction domains.

Explicit architectural biases like modularity can further boost automation efficiency. Nainani (2024) find that models trained with *brain-inspired modular training* (Liu et al., 2023a) produce more readily identifiable circuits compared to standard training. Such domain-inspired inductive biases may prove increasingly vital as models grow more massive and monolithic.

Interpreting Extracted Circuits. Once critical circuit components have been isolated, the key remaining step is interpreting *what* computation those components perform. Sparse autoencoders are a prominent approach for interpreting extracted circuits by decomposing neural network activations into individual component *features*, as discussed in Section 4.2.

A novel paradigm uses large language models themselves as an interpretive tool. Bills et al. (2023) demonstrate generating natural language descriptions of individual neuron functions by prompting language models like GPT-4 to explain sets of inputs that activate a neuron. Mousi et al. (2023) similarly employ language models to annotate unsupervised neuron clusters identified via hierarchical clustering. Bai et al. (2024) describe the roles of neurons in vision networks with multimodal models. These methods can easily leverage more capable general-purpose models in the future. Foote et al. (2023) take a complementary graph-based approach in their neuron-to-graph tool: automatically extracting individual neurons’ behavior patterns from training data as structured graphs amenable to visualization, programmatic comparisons, and property searches. Such representations could synergize with language model-based annotation to provide descriptions of neuron roles.

However, robustly interpreting the largest trillion-parameter models using automated techniques remains an open challenge. Another novel approach, mechanistic-interpretability-based program synthesis (Michaud et al., 2024), entirely sidesteps this complexity by auto-distilling the algorithm learned by a trained model into human-readable Python code without relying on further interpretability analyses or model architectural knowledge. As models become increasingly vast and opaque, such synergistic combinations of methods – uncovering circuits, annotating them, or altogether transcribing them into executable code – will likely prove crucial for maintaining insight and *oversight* when scaling model size.

6 Relevance to AI Safety

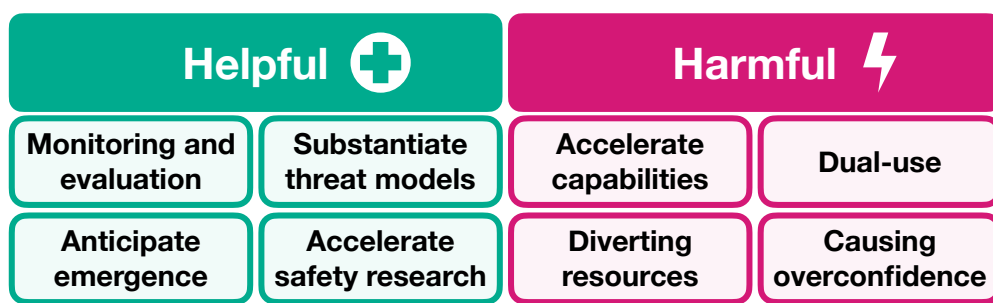


Figure 11: Potential benefits and risks of mechanistic interpretability for AI safety.

How Could Interpretability Promote AI Safety? Gaining mechanistic insights into the inner workings of AI systems seems crucial for navigating AI safety as we develop more powerful models (Nanda, 2022e). Interpretability tools can provide an understanding of artificial cognition, the way AI systems process information and make decisions, which offers several potential benefits:

Mechanistic interpretability could accelerate AI safety research by providing richer feedback loops and grounding for model evaluation (Casper, 2023). It may also help anticipate emergent capabilities, such as the emergence of new skills or behaviors in the model before they fully manifest (Wei et al., 2022; Steinhart, 2023; Nanda et al., 2023a; Barak et al., 2022). This relates to studying the incremental development

of internal structures and representations as the model learns (Section 5.2). Additionally, interpretability could substantiate theoretical risk models with concrete evidence, such as demonstrating *inner misalignment* (when a model’s behavior deviates from its intended goals) or *mesa-optimization* (the emergence of unintended subagents within the model) (Hubinger et al., 2019; von Oswald et al., 2023). It may also trigger normative shifts within the AI community toward rigorous safety protocols by revealing potential risks or concerning behaviors (Hubinger, 2019a).

Regarding specific AI risks (Hendrycks et al., 2023), interpretability may prevent malicious misuse by locating and erasing sensitive information stored in the model (Meng et al., 2022a; Nguyen et al., 2022). It could reduce competitive pressures by substantiating potential threats, promoting organizational safety cultures, and supporting AI alignment (ensuring AI systems pursue intended goals) through better monitoring and evaluation (Hendrycks & Mazeika, 2022). Interpretability can provide safety filters for every stage of training: before training by deliberate design (Hubinger, 2019a), during training by detecting early signs of misalignment and potentially shifting the distribution towards alignment (Hubinger, 2022; Sharkey, 2022), and after training by rigorous evaluation of artificial cognition for honesty (Burns et al., 2023; Zou et al., 2023) and screening for deceptive behaviors (Park et al., 2023b).

The emergence of *internal world models* in LLMs, as posited by the *simulation* hypothesis, could have significant implications for AI alignment research. Finding an internal representation of human values and aiming the AI system’s objective may be a trivial way to achieve alignment (Wentworth, 2022), especially if the world model is internally separated from notions of goals and agency (Ruthenis, 2022). In such cases, world model interpretability alone may be sufficient for alignment (Ruthenis, 2023).

Conditioning pre-trained models is considered a comparatively safe pathway towards general intelligence, as it avoids directly creating agents with inherent goals or agendas (Jozdien, 2022; Hubinger et al., 2023). However, prompting a model to simulate an actual agent, such as "You are a superintelligence in 2035 writing down an alignment solution," could inadvertently lead to the formation of internal agents (Hubinger et al., 2023). In contrast, reinforcement learning tends to create agents by default (Casper et al., 2023a; Ngo et al., 2022).

The *prediction orthogonality* hypothesis suggests that prediction-focused models like GPT can simulate agents with potentially misaligned objectives (janus, 2022). Although GPT may lack genuine agency or intentionality, it may produce outputs that simulate these qualities (Bereska & Gavves, 2023; Shanahan et al., 2023). This underscores the need for careful *oversight* and, better yet, using mechanistic interpretability to search for internal agents or their constituents, such as optimization or search processes – an endeavor known as *searching for search* (NicholasKees & janus, 2022; Jenner et al., 2024).

Mechanistic interpretability integrates well into various AI alignment agendas, such as understanding existing models, controlling them, making AI systems solve alignment problems, and developing alignment theories (technicalities & Stag, 2023; Hubinger, 2020). It could enhance strategies like detecting *deceptive alignment* (hypothetical when a model ensures to appear aligned as to pursue misaligned goals without raising suspicion) (Park et al., 2023b), *eliciting latent knowledge* from models (Christiano et al., 2021), and enabling better scalable *oversight*, such as in *iterative distillation and amplification* (Chan, 2023). A high degree of understanding may even allow for *well-founded AI* approaches (AI systems with provable guarantees) (Tegmark & Omohundro, 2023) or *microscope AI* (extract world knowledge from the model without letting the model take actions) (Hubinger, 2019a). Furthermore, comprehensive interpretability itself may be an alignment strategy if we can identify internal representations of human values and guide the model to pursue those values by *retargeting an internal search process* (Wentworth, 2022). Ultimately, *understanding and control are intertwined*, and deeper understanding can control AI systems more reliably.

However, there is a spectrum of potential misalignment risks, ranging from acute, *model-centric* issues to gradual, *systemic* concerns (Kulveit, 2024). While mechanistic interpretability may address risks stemming directly from model internals – such as deceptive alignment or sudden capability jumps – it may be less helpful for tackling broader systemic risks like the emergence of misaligned economic structures or novel evolutionary dynamics (Hendrycks, 2023b). The multi-scale risk landscape calls for a balanced research portfolio to minimize risk, where research on governance, complex systems, and multi-agent simulations complements